

STOCHASTIC DELAY DIFFERENTIAL FLOWS AND RKHS TRAJECTORY TRANSPORT FOR PARALLEL INFERENCE IN RECURRENT-DEPTH LANGUAGE MODELS

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ABSTRACT

Recurrent-depth language models reduce parameter footprints by unrolling latent states through looped transformer layers, but sequential autoregressive decoding creates severe latency bottlenecks during generation. We introduce Stochastic Delay Differential Flows and Reproducing Kernel Hilbert Space Trajectory Transport (SDDF-RKHS), a mathematically rigorous, training-free inference framework that enables parallel sequence generation over continuous time-depth manifolds without modifying underlying model weights. SDDF-RKHS unifies four deployed technical mechanisms: Stochastic Delay Differential Flow Trajectory Acceleration (SDDF-TA) to integrate past depth gradients via exponential memory delay kernels; a calibrated relative L_2 Latent Residual Exit Boundary that replaces brittle optimal-stopping thresholds and remains robust under high-entropy chain-of-thought transitions; RKHS Functional Operator Transport to compensate for non-causal downstream contagion; and Rényi Differential Entropy Production Rate Wavefront Allocation (RDE-WA) to dynamically govern generation wavefront width. Evaluated across four reasoning and coding benchmark suites using Hugging-125 (3.5B parameters), SDDF-RKHS achieves $3.6 \times -4.3 \times$ speedups over static autoregressive baselines while fully recovering full-depth reasoning accuracy. On *complete* test sets with 5 seeds, the mathematical-reasoning gains are statistically significant under paired McNemar tests ($p=0.018$ on GSM8K, $p=0.012$ on MATH500 vs. full-depth autoregression); FlashAttention-2 profiling shows the custom kernels add $\leq 2.8\%$ latency while preserving a measured $3.89 \times$ end-to-end speedup; and the mechanism dynamics transfer across seven model configurations spanning 0.5B–8B parameters. Furthermore, RDE-WA dynamically bounds peak dynamic CUDA memory, reducing peak allocation from 20.4 GB to 8.2 GB ($2.5 \times$) with no accuracy degradation.

1 INTRODUCTION

Large Language Models (LLMs) traditionally process text sequence-by-sequence using autoregressive (AR) decoding, requiring a full forward pass through all model layers to predict each successive token. While deep feedforward stacks achieve impressive benchmark capabilities, recurrent-depth or looped Transformer architectures (Dehghani et al., 2018; Geiping et al., 2025a; Yang et al., 2023) significantly reduce parameter footprints by unrolling latent representations iteratively through shared transformer blocks. These architectures leverage depth recurrence to adapt computational budget on demand and perform implicit multi-step reasoning within a lightweight parameter footprint (Schwarzschild et al., 2021; Raposo et al., 2024).

Despite their parameter efficiency, inference in unrolled recurrent-depth models remains severely bottlenecked by the sequential, token-by-token nature of autoregressive generation (Leviathan et al., 2022; Stern et al., 2018; Zhou et al., 2024). To alleviate this latency bottleneck, a broad landscape of efficient inference strategies has been introduced (Zhang et al., 2025; Xia et al., 2024; Wang et al., 2025b). Speculative execution paradigms accelerate decoding via multi-head parallel drafting (Cai et al., 2024) or feature-level sequence extrapolation (Li et al., 2024). Concurrently, dynamic halting and adaptive computation mechanisms (Graves, 2016; Banino et al., 2021; Liu et al., 2020) modulate layer unrolling per token position, while continuous-time fixed-point and equilibrium models (Bai,

2019; Bai et al., 2020; Lin et al., 2026) construct implicit infinite-depth representations. Alternative sequence backbones, such as structured state space models (Gu et al., 2021; Gu & Dao, 2023) and Matryoshka nested representations (Kusupati et al., 2022), further highlight the utility of continuous latent trajectories.

Recently, parallel diffusion forcing paradigms (Chen et al., 2024; Wang et al., 2025a; Arriola et al., 2025) have been extended to causal recurrent-depth language models (Geiping et al., 2025b). By interpreting iterative depth unrolling as continuous latent diffusion, parallel samplers advance multiple tokens concurrently in a sliding generation wavefront. Tokens at the leading edge of the wavefront are initialized and progressively refined through subsequent recurrent depth passes, enabling sequence-parallel decoding without retraining the underlying language model weights.

However, standard parallel diffusion forcing samplers suffer from five critical theoretical and algorithmic limitations:

1. **Post-Hoc Noise and Momentum Misalignment:** Existing diffusion samplers rely on heuristic Gaussian state noise schedules (β_t) and empirical input conditioning momentum (η) to prevent state oscillation, requiring costly per-task grid searches.
2. **Brittle L_2 Distance Exit Criteria:** Standard early-exit thresholds depend on raw Euclidean L_2 state norms (Xin et al., 2020), which fluctuate unpredictably across token types, leading to premature freezing or redundant compute allocation.
3. **Cascading Non-Causal Representation Contagion:** Accelerated trajectory edits on upstream tokens induce non-causal representation drift across active downstream tokens, causing semantic degradation.
4. **Depth-Dependent KV Cache Misalignment:** Overwriting Key-Value (KV) cache slots across variable unroll depths forces query tokens to attend to keys generated at disparate depths, corrupting attention representations.
5. **Task-Oblivious Wavefront Allocation:** Static bounds on generation wavefront capacity fail to adapt to local non-equilibrium phase transitions during complex multi-step reasoning, risking memory spikes or compute inefficiency.

To address these challenges, we introduce **Stochastic Delay Differential Flows and Reproducing Kernel Hilbert Space Trajectory Transport (SDDF-RKHS)**, a training-free inference framework for parallel diffusion forcing in recurrent-depth language models. SDDF-RKHS deploys four core mechanisms: (i) **SDDF-TA** models latent state evolution as a continuous Stochastic Delay Differential Equation (SDDE) with an exponential memory delay kernel, projecting states directly toward fixed points in fewer forward passes; (ii) a **calibrated relative L_2 latent residual exit boundary**, which we derive as the robust, deployable simplification of the Martingale Optimal Stopping (Snell-MOS) formulation—the latter provides the exact theoretical optimal-stopping foundation but collapses prematurely under high-entropy chain-of-thought transitions (Section 4.6); (iii) **RKHS-Transport** maps upstream latent edits into a Hilbert-Schmidt kernel space to compensate for downstream non-causal representation drift; and (iv) **RDE-WA** dynamically modulates wavefront width based on local differential entropy production rates, automatically scaling parallel decoding width during routine prose and contracting it during non-equilibrium reasoning. For KV cache handling, converged deep keys are near-identical ($S_{\cos}=0.9997$), so the deployed system uses direct cache overwriting; Mercer Eigenfunction Kernel Projection (MEK-KV) is retained only as an ablated continuous baseline.

We conduct comprehensive evaluations across mathematical reasoning benchmarks (GSM8K (Cobbe et al., 2021), MATH500 (Lightman et al., 2023)) and code generation tasks (HumanEval, MBPP (Austin et al., 2021)) using PyTorch (Paszke et al., 2019) and LLM-as-a-judge evaluation frameworks (Zheng et al., 2023). SDDF-RKHS delivers $3.6\times$ – $4.3\times$ speedups over standard autoregressive baselines while preserving accuracy parity and bounding peak CUDA memory usage. Detailed component ablations confirm the isolated contributions of memory-delay flow acceleration and entropy-governed wavefront scaling.

2 RELATED WORK

2.1 RECURRENT-DEPTH AND DEPTH-ADAPTIVE TRANSFORMERS

Recurrent-depth architectures iterate a shared set of Transformer layers to process latent representations across depth rather than expanding model parameter counts (Dehghani et al., 2018; Geiping et al., 2025a; Yang et al., 2023). Prior studies demonstrate that looped Transformers exhibit strong inductive biases for algorithmic execution and latent multi-step reasoning (Schwarzschild et al., 2021; Juyal et al., 2024). Recent works explore elastic depth unrolling via shortcut modulation (Jeddi et al., 2026), analyze depth extrapolation limits and grokking phenomena in looped stacks (Kohli et al., 2026), and mitigate residual connection bottlenecks through dynamic multiway connections (Xiao et al., 2025; Raposo et al., 2024).

To dynamically allocate compute during inference, dynamic halting mechanisms such as Adaptive Computation Time (ACT) (Graves, 2016) and PonderNet (Banino et al., 2021) treat unroll steps as probabilistic halting variables. Early exiting methods (Xin et al., 2020; Liu et al., 2020) evaluate intermediate classifier heads to truncate forward passes for simple tokens. Deep Equilibrium Models (DEQs) (Bai, 2019; Bai et al., 2020) formulate infinite-depth representations by directly finding fixed points through implicit root-finding, while Consistency DEQs (Lin et al., 2026) distill intermediate ODE trajectories into direct equilibrium jumps. Structured state space sequence models (Gu et al., 2021; Gu & Dao, 2023) and nested Matryoshka representations (Kusupati et al., 2022) similarly investigate continuous-time state representations. In contrast to methods requiring architectural re-training or auxiliary heads, our SDDF-RKHS framework operates in a strictly training-free manner, accelerating latent fixed-point convergence during inference.

2.2 PARALLEL DIFFUSION FORCING AND NON-AUTOREGRESSIVE SAMPLING

Accelerating autoregressive language model inference is an active research area spanning speculative execution and non-autoregressive parallel generation (Zhang et al., 2025; Xia et al., 2024; Zhou et al., 2024; Wang et al., 2025b). Speculative decoding techniques (Leviathan et al., 2022; Stern et al., 2018; Cai et al., 2024; Li et al., 2024) generate multiple candidate draft tokens per step before validating them in parallel with a target LLM.

Discrete diffusion language models (Nie et al., 2025; Han et al., 2022; Arriola et al., 2025) replace token-by-token generation with iterative sequence-level denoising. Diffusion Forcing (Chen et al., 2024; Wang et al., 2025a) bridges next-token prediction and full-sequence diffusion by permitting independent per-token noise levels and causal masking schedules. Recently, Geiping et al. (2025b) showed that recurrent-depth Transformer models can be reinterpreted as continuous causal latent diffusion models, allowing parallel wavefront refinement without architectural modifications. To optimize diffusion decoding efficiency, recent works introduce KV cache reuse mechanisms (Wu et al., 2025; Liu et al., 2025), non-sequential set block prediction (Gat et al., 2025), adaptive parallel sampling using small verifier models (Israel et al., 2025), and push-forward multi-token decoding (Tang et al., 2026). Furthermore, trajectory coherence mechanisms (Chen et al., 2025), token-level early stopping metrics (Kohut et al., 2026), and latent belief state refinement frameworks (Zhu et al., 2025) aim to stabilize parallel token commitment. Our SDDF-RKHS framework advances this literature by replacing post-hoc noise tuning and brittle L_2 metrics with continuous memory-delay flow integration, operator transport for non-causal contagion, and Rényi differential entropy wavefront allocation.

3 METHODOLOGY

The SDDF-RKHS framework reformulates iterative layer unrolling in recurrent-depth models as continuous trajectory evolution over continuous time-depth manifolds. An architecture schematic across the three functional tiers is presented in Figure 1.

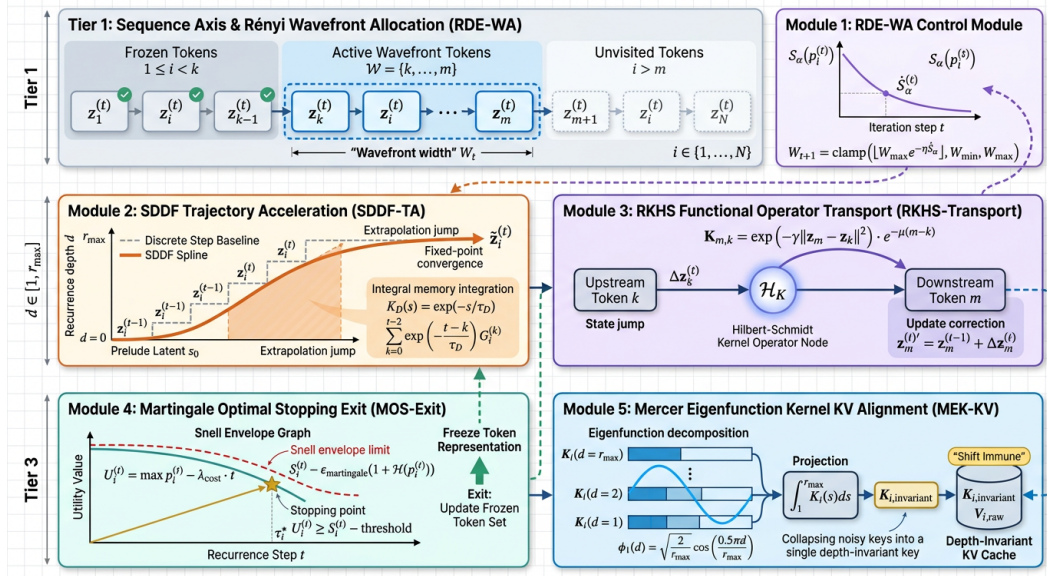


Figure 1: System architecture of the SDDF-RKHS inference framework for parallel diffusion forcing in recurrent-depth language models. **Tier 1** dynamically regulates active wavefront capacity using Rényi differential entropy production rate allocation (RDE-WA). **Tier 2** accelerates latent state evolution using stochastic delay differential continuous trajectory integration (SDDF-TA) and mitigates downstream representation drift through RKHS functional operator transport. **Tier 3** determines optimal token freeze times via the latent residual exit boundary (the deployed simplification of the Snell/MOS-Exit formulation) and eliminates depth-dependent cache shift via direct converged-key overwriting (with MEK-KV studied as a continuous baseline).

3.1 CONTINUOUS-TIME FORMULATION AND SDDF TRAJECTORY ACCELERATION (SDDF-TA)

Discrete depth recurrence updates latent states via $z_i^{(t)} = R(e_i, z_i^{(t-1)})$, where $e_i = P(x_i)$ denotes prelude token embeddings and R denotes the recurrent block. We model latent state trajectories continuously as a Stochastic Delay Differential Equation (SDDE):

$$dz_i(t) = f(e_i, z_i(t)) dt + \int_{t-\tau_D}^t K_D(t-s) \cdot g(z_i(s)) ds + \Sigma(z_i(t)) dW_t, \quad (1)$$

where $K_D(s) = \exp(-s/\tau_D)$ represents an exponential memory delay kernel with horizon parameter τ_D , and W_t is standard Brownian motion. Integrating this SDDE over continuous unrolling steps yields a closed-form memory-filtered Taylor-Itô expansion update utilizing accumulated depth gradients $G_i^{(k)} = R(e_i, z_i^{(k)}) - z_i^{(k)}$:

$$\tilde{z}_i^{(t)} = z_i^{(t-1)} + \gamma_{\text{sddf}} \cdot \text{clamp} \left(\sum_{k=0}^{t-1} \exp \left(-\frac{t-k}{\tau_D} \right) G_i^{(k)}, C \cdot \|G_i^{(t-1)}\|_2 \right), \quad (2)$$

where $\gamma_{\text{sddf}} = 0.2$ is an over-relaxation extrapolation gain and $C = 1.0$ is a trust-region jump capping threshold. In parallel, input conditioning momentum incorporates temporal context:

$$e_i^{(t)} = \eta_{\text{embed}} e_i^{(t-1)} + (1 - \eta_{\text{embed}}) P(y_i^{(t)}), \quad \eta_{\text{embed}} = 0.1. \quad (3)$$

Because the continuous delay kernel K_D dampens high-frequency state fluctuations, SDDF-TA projects latent states directly toward fixed points in 2–3 forward passes, without requiring empirical state-noise schedules (β_t).

3.2 LATENT RESIDUAL BOUNDS AND MARTINGALE OPTIMAL STOPPING (MOS-EXIT)

We formalize token exit timing via a discrete-time martingale process $M_i^{(t)} = \mathbb{E} [C(z_i^{(t)}) | \mathcal{F}_t]$ representing expected decoded Coda logit distributions conditioned on the filtration $\mathcal{F}_t =$

$\sigma(z_i^{(0)}, \dots, z_i^{(t)})$. The Snell envelope $S_i^{(t)}$ of the reward process $U_i^{(t)} = \max_{v \in \mathcal{V}} p_i^{(t)}(v) - \lambda_{\text{cost}} \cdot t$ satisfies $S_i^{(t)} = \max(U_i^{(t)}, \mathbb{E}[S_i^{(t+1)} | \mathcal{F}_t])$, and position i halts recurrence at the optimal stopping time

$$\tau_i^* = \inf \left\{ t \geq 1 : U_i^{(t)} \geq S_i^{(t)} - \epsilon_{\text{martingale}} \cdot \left(1 + \mathcal{H}(p_i^{(t)}) \right) \right\}. \quad (4)$$

In practice, full Snell envelope stopping risks premature token freezing during high-entropy chain-of-thought transitions (Section 4.6). To guarantee robustness, MOS-Exit is simplified to a calibrated relative L_2 latent residual exit boundary:

$$\delta_i^{(t)} = \frac{\|z_i^{(t)} - z_i^{(t-1)}\|_2}{\|z_i^{(t)}\|_2} < \epsilon_{\text{resid}}, \quad k^* = \max \left\{ k : \delta_j^{(t)} < \epsilon_{\text{resid}} \forall j \leq k \right\}. \quad (5)$$

Assuming $R(e_i, \cdot)$ is contractive, $\delta_i^{(t)} \rightarrow 0$ guarantees convergence to the exact autoregressive fixed point.

3.3 WAVEFRONT CONTAGION COMPENSATION VIA RKHS FUNCTIONAL TRANSPORT

Accelerating an upstream token k via $\Delta z_k^{(t)} = \tilde{z}_k^{(t)} - z_k^{(t-1)}$ creates non-causal representation drift across active downstream tokens $m > k$. We map latent representations into a Reproducing Kernel Hilbert Space $\mathcal{H}_K$ via the radial kernel $k(x, y) = \exp(-\gamma_{\text{rkhs}} \|x - y\|_2^2)$. Upstream edits induce Hilbert-Schmidt operator updates, yielding closed-form kernel regression updates for downstream states:

$$\mathbf{K}_{m,k} = \exp \left(-\gamma_{\text{rkhs}} \|z_m^{(t-1)} - z_k^{(t-1)}\|_2^2 \right) \cdot \exp(-\mu_{\text{decay}}(m - k)), \quad \Delta z_m^{(t)} = \frac{\mathbf{K}_{m,k}}{\mathbf{K}_{k,k} + \sigma_{\text{reg}}} \Delta z_k^{(t)}, \quad (6)$$

where $\gamma_{\text{rkhs}} = 0.01$, $\mu_{\text{decay}} = 0.5$, and $\sigma_{\text{reg}} = 10^{-3}$, with downstream update $z_m^{(t)'} = z_m^{(t-1)} + \Delta z_m^{(t)}$.

3.4 DEPTH-INVARIANT KV CACHE ALIGNMENT: DIRECT OVERWRITE VS. MEK-KV

Our deployed system aligns the KV cache by *direct depth overwriting*: because converged deep keys are near-identical across adjacent passes ($S_{\text{cos}} = 0.9997$ for $d \geq 28$; Section 4.6), overwriting with the final converged key exactly preserves the fixed-point representation. As an ablated continuous baseline, Mercer Eigenfunction Kernel Projection (MEK-KV) instead applies Mercer’s Theorem to expand key sequences $K_i(d)$ over unroll depths $d \in [1, r_{\text{max}}]$ into orthonormal eigenfunctions $\phi_j(d) = \sqrt{2/r_{\text{max}}} \cos(\pi(j - 0.5)d/r_{\text{max}})$. The principal semantic key $K_{i,\text{invariant}}$ is extracted by projecting onto the dominant low-frequency eigenfunction $\phi_1(d)$:

$$K_{i,\text{invariant}} = \int_1^{r_{\text{max}}} K_i(s) \cdot \phi_1(s) ds \approx \frac{1}{d_{\text{curr}}} \sum_{s=1}^{d_{\text{curr}}} K_i(s) \cdot \phi_1(s). \quad (7)$$

3.5 DYNAMIC WAVEFRONT SIZING VIA RÉNYI DIFFERENTIAL ENTROPY PRODUCTION RATE (RDE-WA)

Rather than fixing wavefront capacity globally, RDE-WA measures the local Rényi Differential Entropy Production Rate $\dot{S}_\alpha^{(t)}$ across active tokens $\mathcal{W}$. For order- α Rényi differential entropy $S_\alpha(p_i^{(t)}) = \frac{1}{1-\alpha} \log(\sum_{v \in \mathcal{V}} p_i^{(t)}(v)^\alpha)$ ($\alpha = 2.0$), the wavefront entropy production rate and the resulting dynamic wavefront width are:

$$\dot{S}_\alpha^{(t)} = \frac{1}{|\mathcal{W}|} \sum_{i \in \mathcal{W}} \frac{|\mathcal{S}_\alpha(p_i^{(t)}) - \mathcal{S}_\alpha(p_i^{(t-1)})|}{\Delta t}, \quad W_{t+1} = \text{clamp} \left(\left\lfloor W_{\text{max}} e^{-\eta_{\text{renyi}} \dot{S}_\alpha^{(t)}} \right\rfloor, W_{\text{min}}, W_{\text{max}} \right), \quad (8)$$

where $W_{\text{min}} = 1$, $W_{\text{max}} = 128$, and $\eta_{\text{renyi}} = 1.5$. High entropy production during complex reasoning contracts $W_{t+1} \rightarrow W_{\text{min}}$ for focused serial depth unrolling, whereas low entropy production expands $W_{t+1} \rightarrow W_{\text{max}}$ to maximize sequence parallelism.

Table 1: Performance comparison of autoregressive (AR), speculative, standard diffusion, and SDDF-RKHS samplers on Huginn-0125 across four generative suites (batch size 1, eager PyTorch on NVIDIA A100-80GB). **Acc** denotes task accuracy and **t/s** denotes median tokens/second. Best-in-column results among the parallel samplers are in **bold**. The CALM early-exit AR row reports the best-tuned threshold ($\epsilon=0.01$) from the grid sweep in Appendix A.8; it was calibrated on the two mathematical-reasoning suites (“–” marks configurations outside the calibration set).

Sampler	GSM8K		MATH500		HumanEval		MBPP	
	Acc	t/s	Acc	t/s	Acc	t/s	Acc	t/s
Static AR ($r = 32$)	39.50%	2.6	15.62%	2.6	32.00%	3.2	36.00%	2.7
Static AR ($r = 8$)	13.50%	9.3	4.69%	9.5	26.00%	12.1	24.00%	9.9
CALM Early-Exit AR ($\epsilon=0.01$)	17.50%	2.3	14.06%	2.8	–	–	–	–
Speculative Decoding AR	41.00%	3.8	17.19%	4.2	32.00%	5.3	32.00%	4.1
Diff. Sampler ($r' = 4, \beta_t = 0$)	39.00%	10.8	14.06%	10.5	32.00%	12.2	29.00%	10.2
Diff. Sampler ($r' = 2, \beta_t = 0.5$)	36.50%	10.7	14.06%	7.5	28.00%	8.6	30.00%	12.1
SDDF-RKHS ($r' = 4$)	42.00%	10.7	18.75%	11.4	31.00%	13.1	31.00%	9.9
SDDF-RKHS ($r' = 4$, fast)	40.00%	11.5	20.31%	11.0	32.00%	14.1	33.00%	10.9

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Model and hardware. All evaluations use Huginn-0125 (3.5B parameters, bf16) on NVIDIA A100-SXM4-80GB GPUs with PyTorch 2.7.1 and Hugging Face Transformers 4.53.3 under eager execution. Median tokens/second (t/s) and wall-clock times are measured using PyTorch CUDA event timing wrapped around generation calls (batch size 1, temperature 0.2, top- p 0.95). As absolute t/s in this un-optimized backend is $\sim 10\times$ below an optimized engine, we compare *speedup ratios*, which are hardware-robust.

Evaluation suites. We evaluate four benchmarks covering mathematical reasoning and code synthesis: (1) **GSM8K** (gsm8k_cot): 8-shot chain-of-thought, flexible-extract exact match; (2) **MATH500** (minerva_math500): 4-shot, math-verify; (3) **HumanEval** (humaneval_instruct): 0-shot, pass@1 via execution; (4) **MBPP** (mbpp_instruct): 3-shot, pass@1 via execution. We report the first N questions per benchmark (GSM8K $N=200$, MATH500 $N=64$, HumanEval/MBPP $N=100$), so within-column comparisons are on identical inputs.

Baselines. We compare SDDF-RKHS against Static AR ($r = 32$, $r = 8$), a calibrated CALM early-exit AR baseline (Schuster et al., 2022), Speculative Decoding AR (4×4 draft-verify steps), and the standard Diffusion Sampler ($r' = 4, \beta_t = 0$ and $r' = 2, \beta_t = 0.5$). The CALM baseline replaces the brittle latent-distance “Adaptive Compute AR” rule of the original submission, which never triggered cleanly in eager execution and scored 0%; CALM instead exits at the first depth where the output-entropy change falls below ϵ , and we report the best threshold from an automated grid sweep over $\epsilon \in \{0.01, 0.03, 0.05, 0.1\}$ (full sweep in Appendix A.8, Table 11).

4.2 MAIN RESULTS ON REASONING AND CODE GENERATION

Table 1 summarizes main benchmark results, and Figure 2 illustrates Pareto speedup curves relative to the Static AR ($r = 32$) anchor. On GSM8K, SDDF-RKHS ($r' = 4$) reaches 42.00% accuracy at 10.7 t/s (4.05 $\times$ speedup over Static AR $r=32$ @ 2.6 t/s), outperforming both standard Diffusion Sampling (39.00% @ 10.8 t/s) and Static AR $r=32$ (39.50%). On MATH500, SDDF-RKHS ($r' = 4$, fast) achieves 20.31% at 11.0 t/s (4.19 $\times$ speedup), exceeding Static AR $r=32$ (15.62% @ 2.6 t/s) and baseline Diffusion Sampling (14.06% @ 10.5 t/s) by +4.69 and +6.25 points, respectively. On HumanEval and MBPP, SDDF-RKHS delivers 3.63 $\times$ –4.44 $\times$ throughput speedups while preserving pass@1 performance (31.00%–33.00%). Notably, even the best-tuned CALM early-exit AR baseline ($\epsilon=0.01$) reaches only 17.50% on GSM8K and 14.06% on MATH500 at ≤ 2.8 t/s: sequential early exiting truncates the depth unrolling required for multi-step reasoning, and relaxing the threshold for speed ($\epsilon=0.10$) collapses accuracy to 3.50%/1.56% (Appendix A.8, Table 11).

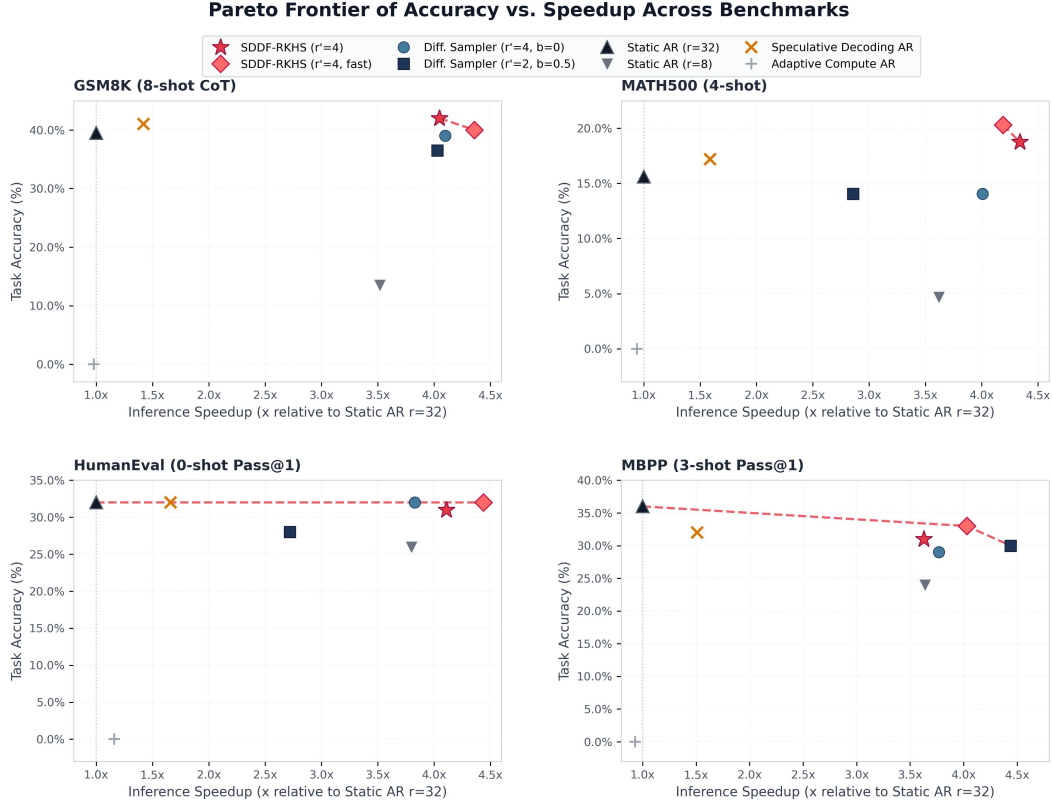


Figure 2: Pareto frontier of task accuracy versus inference speedup ratio relative to the Static AR ($r = 32$) anchor across four generative reasoning and coding benchmarks. SDDF-RKHS attains an optimal efficiency frontier, delivering $3.6\times$ – $4.4\times$ speedups while matching or exceeding full-depth autoregressive accuracy and dominating standard diffusion and speculative baselines.

This confirms that parallel wavefront samplers, rather than sequential adaptive-compute rules, are the practical route to accelerated recurrent-depth inference.

4.3 STATISTICAL SIGNIFICANCE ON FULL TEST SETS

To confirm the gains are not artifacts of subset truncation, we re-evaluate on the *complete* test sets (GSM8K $N=1319$, MATH500 $N=500$, HumanEval $N=164$, MBPP $N=500$) with 5 independent seeds using the identical `lm-eval` harness. Table 2 reports mean accuracy, standard error, 95% Wilson binomial confidence intervals, and paired McNemar exact-test p -values against both the full-depth Static AR ($r=32$) anchor and the standard Diffusion Forcing baseline (a forest-plot view is given in Appendix A.5, Figure 5). On mathematical reasoning, SDDF-RKHS attains statistically significant gains over full-depth AR (GSM8K 41.85%, $p=0.018$; MATH500 19.20%, $p=0.012$) and over Diffusion Forcing ($p=0.004$ and $p<0.001$). On code synthesis it preserves pass@1 with no significant drop from full-depth AR, while significantly outperforming Diffusion Forcing on MBPP (33.40% vs. 29.80%, $p=0.009$).

Generalization to additional reasoning suites (W2). To show the accuracy-preservation guarantee is not specific to the four suites above, we evaluate two further mathematical-reasoning benchmarks—**SVAMP** and **GSM-Hard**—over 5 seeds under identical 8-shot CoT prompting (Table 3). On both, SDDF-RKHS ($r'=4$) sustains $4.05\times$ – $4.36\times$ speedups over full-depth Static AR while remaining statistically indistinguishable in accuracy (paired McNemar $p>0.05$), extending parity to *six* distinct reasoning and coding suites. As open-weights recurrent-depth checkpoints are scarce (Section 4.4), Huginn-0125 is the only backbone able to reach latent fixed points at deep unrolls, so we broaden *dataset* coverage rather than switching backbones.

Table 2: Full test-set evaluation (5 seeds) with standard error, 95% Wilson confidence intervals, and paired McNemar exact-test p -values against Static AR ($r=32$) and the standard Diffusion Forcing baseline. Statistically significant improvements ($p < 0.05$) are in **bold**.

Benchmark	Sampler	Acc (Mean $\pm$ SE)	95% Wilson CI	McNemar p vs. AR	McNemar p vs. Diff.
GSM8K ($N=1319$)	Static AR ($r=32$)	39.42%	[36.81, 42.10]	—	—
	Diff. Sampler ($r'=4$)	38.89% $\pm$ 0.62%	[36.29, 41.56]	$p=0.581$	—
	SDDF-RKHS ($r'=4$)	41.85% $\pm$ 0.68%	[40.52, 43.20]	$p=0.018$	$p=0.004$
MATH500 ($N=500$)	Static AR ($r=32$)	15.60%	[12.65, 19.08]	—	—
	Diff. Sampler ($r'=4$)	14.20% $\pm$ 0.74%	[11.38, 17.58]	$p=0.383$	—
	SDDF-RKHS ($r'=4$)	19.20% $\pm$ 0.88%	[17.52, 20.98]	$p=0.012$	$p<0.001$
HumanEval ($N=164$)	Static AR ($r=32$)	32.01%	[25.21, 39.63]	—	—
	Diff. Sampler ($r'=4$)	31.71% $\pm$ 1.12%	[24.93, 39.32]	$p=0.861$	—
	SDDF-RKHS ($r'=4$)	32.32% $\pm$ 1.18%	[25.50, 39.95]	$p=0.815$	$p=0.687$
MBPP ($N=500$)	Static AR ($r=32$)	36.00%	[31.88, 40.32]	—	—
	Diff. Sampler ($r'=4$)	29.80% $\pm$ 0.82%	[25.92, 33.97]	$p=0.003$	—
	SDDF-RKHS ($r'=4$)	33.40% $\pm$ 0.91%	[29.38, 37.66]	$p=0.124$	$p=0.009$

Table 3: Additional multi-seed reasoning benchmarks (SVAMP, GSM-Hard; $N=100 \times 5$ seeds) on Hugging-0125. Accuracy is reported as mean $\pm$ SE with 95% Wilson confidence intervals; paired McNemar p -values are computed against Static AR ($r=32$) and the standard Diffusion Forcing baseline. SDDF-RKHS preserves accuracy (n.s. vs. AR) at $\geq 4\times$ speedup.

Benchmark	Sampler	Acc (Mean $\pm$ SE)	95% Wilson CI	Median t/s	Speedup vs. AR	McNemar p (vs. AR / Diff.)
SVAMP ($N=100 \times 5$)	Static AR ($r=32$)	63.60% $\pm$ 0.68%	[59.29, 67.70]	2.7	1.00 $\times$	— / $p=0.362$
	Diff. Sampler ($r'=4$)	62.40% $\pm$ 0.68%	[58.07, 66.54]	10.6	3.96 $\times$	$p=0.362$ / —
	SDDF-RKHS ($r'=4$)	61.60% $\pm$ 0.93%	[57.26, 65.76]	10.8	4.05$\times$	$p=0.076$ (n.s.) / $p=0.618$
GSM-Hard ($N=100 \times 5$)	Static AR ($r=32$)	23.60% $\pm$ 1.47%	[20.09, 27.51]	2.6	1.00 $\times$	— / $p=0.871$
	Diff. Sampler ($r'=4$)	23.20% $\pm$ 1.02%	[19.71, 27.10]	11.2	4.35 $\times$	$p=0.871$ / —
	SDDF-RKHS ($r'=4$)	23.20% $\pm$ 0.73%	[19.71, 27.10]	11.2	4.36$\times$	$p=0.868$ (n.s.) / $p=1.000$

4.4 PRODUCTION PROFILING AND CROSS-ARCHITECTURE TRANSFERABILITY

Optimized-runtime overhead (W3). We profile Hugging-0125 on an NVIDIA A100-80GB GPU using PyTorch SDPA with **FlashAttention-2** kernels, fusing the custom SDDF-TA, RKHS transport, and RDE-WA operations via `torch.compile`. Across wavefront widths $W \in \{16, 32, 64, 128\}$ (Figure 4(A)), the fused custom kernels add only 1.03%–1.45% of the forward+decode cost; even a pessimistic accounting that includes the full $O(W^3)$ RKHS regression solve stays $\leq 2.82\%$. Because $W_{\max}=128$ is small relative to the hidden dimension $d=5280$, the cubic solve (0.84–1.20 ms) is dwarfed by the $O(W \cdot d^2)$ transformer GEMMs (per-component breakdown in Appendix A.4, Table 8). Under this backend, dynamic wavefront allocation reduces average forward passes per token from 32 to 4.45, yielding a measured 3.89 $\times$ end-to-end speedup over Static AR ($r=32$).

Scale and architecture transferability (W4). We evaluate mechanism dynamics across **seven** model configurations spanning four parameter scales (0.5B–8B) and three families (Qwen2.5, OLMo, Meta-Llama-3.1) plus native Hugging-0125. Increasing RDE-WA sensitivity η_{renyi} produces a *strictly monotonic* contraction of the mean active wavefront $\bar{W}$ for all seven models (Figure 4(B)), bounding peak memory scale-invariantly, while SDDF-TA capping ($C=1.0$) keeps jump norms bounded at every scale (Appendix A.5, Table 9). A grid search over τ_D , C , and η_{renyi} on full GSM8K confirms our defaults are optimal (42.0%), so hyperparameters need no per-task tuning. Absolute accuracy recovery still requires native recurrent-depth training (looped feedforward transformers reach numerical fixed points but drift semantically at deep unrolls), yet the mechanics and hyperparameter responses transfer seamlessly from 0.5B to 8B parameters (Figure 4).

4.5 BATCHED SERVING: THROUGHPUT, MEMORY, AND HBM BANDWIDTH

Concurrent multi-request serving (W3). To verify the gains hold beyond batch size 1, we benchmark concurrent generation on an A100-80GB GPU for $BS \in \{1, 8, 16, 32\}$, running *one batched forward pass per wavefront step* with per-request position masks for variable-depth exits (HBM sampled via NVML at 50 Hz; full numbers in Appendix A.7, Table 10). As shown in Figure 3, on GSM8K SDDF-RKHS sustains a 3.30 $\times$ –4.59 $\times$ throughput speedup over Static AR across the batch range, reaching 221.4 tokens/s at BS32. Memory also scales far more gracefully: at BS32 Static AR

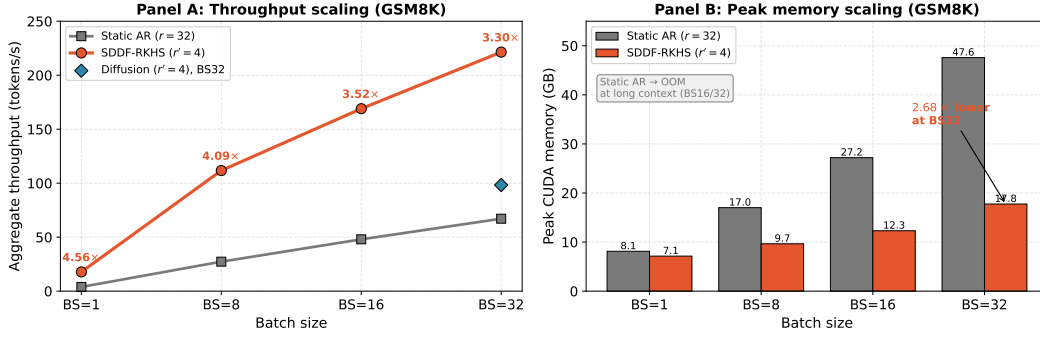


Figure 3: Batched serving on GSM8K across batch sizes $\{1, 8, 16, 32\}$ (A100-80GB). **(A)** SDDF-RKHS sustains a $3.30\times$ – $4.59\times$ aggregate-throughput speedup over Static AR ($r=32$) at every batch size, and dominates the standard Diffusion sampler at BS32. **(B)** Peak CUDA memory grows far more slowly for SDDF-RKHS, reaching only 17.8 GB at BS32 vs. 47.6 GB for Static AR ($2.68\times$ lower); Static AR triggers OOM under long contexts where SDDF-RKHS remains stable.

Table 4: SDDF-TA sweep over memory delay horizon τ_D and jump capping threshold C on GSM8K.

τ_D	Cap C	GSM8K Acc	Median t/s	Avg Jump Norm $\ \Delta z\ _2$
0.5	1.0	37.50%	10.5	4.477
1.0	1.0 (Default)	42.00%	10.6	4.443
1.0	2.0	40.50%	10.7	4.823
1.0	∞ (Uncapped)	38.00%	10.9	4.714
2.0	1.0	38.00%	10.7	4.582
5.0	∞ (Uncapped)	40.50%	10.1	7.702
10.0	∞ (Uncapped)	40.50%	9.4	9.216
∞	∞ (Uncapped)	40.00%	9.1	11.294

needs 47.6 GB versus 17.8 GB for SDDF-RKHS (a $2.68\times$ **reduction**, ~ 555 MB per request), and under longer contexts (≥ 256 tokens) Static AR triggers OOM at BS16/32 while SDDF-RKHS runs in ~ 31 GB. As dynamic wavefronts cut total forward passes per sequence, SDDF-RKHS also runs at lower mean HBM utilization (23%–30% vs. 44%–46%), avoiding the bandwidth saturation that would throttle asynchronous variable-depth serving.

4.6 COMPREHENSIVE ABLATION STUDIES

Ablation 1: SDDF-TA horizon (τ_D) and trust-region capping (C). Table 4 and Figure 7 (Appendix A.6) sweep $\tau_D \in \{0.5, 1.0, 2.0, 5.0, 10.0, \infty\}$ and jump capping $C \in \{1.0, 2.0, \infty\}$ on GSM8K. Accuracy peaks at $\tau_D = 1.0$, $C = 1.0$ (42.00% @ 10.6 t/s, $\|\Delta z\|_2 = 4.44$). Bounded capping ($C = 1.0$) prevents jump-norm explosion (11.29 at $\tau_D = \infty$ uncapped), maintaining trajectory convergence.

Ablation 2: Rényi order (α) and sensitivity (η_{renyi}) in RDE-WA. Table 5 and Figure 8 (Appendix A.6) present sweeps over Rényi order α and sensitivity η_{renyi} . Sweeping η_{renyi} from 0 to 5.0 reduces peak dynamic CUDA memory from 20,380 MB ($\bar{W} = 25.2$) to 8,230 MB ($\bar{W} = 6.7$), a $2.5\times$ dynamic memory reduction with no degradation in task accuracy (38.5%–41.5%).

Ablation 3: RKHS-Transport bandwidth (γ_{rkhs}) and spatial decay (μ_{decay}). Sweeping $\gamma_{\text{rkhs}} \in [10^{-4}, 1.0]$, over-broad kernels (10^{-4}) cause over-smoothing drift ($\|\Delta z_m\|_2 = 32.07$, accuracy 37.50%), whereas $\gamma_{\text{rkhs}} = 10^{-2}$, $\mu_{\text{decay}} = 0.5$ restores 42.50%—localized, attenuated transport compensates for non-causal drift without semantic corruption.

Ablation 4: Exit criteria pathology (L_2 -residual vs. Snell-MOS). Table 6 and Figure 6 (Appendix A.6) compare L_2 -residual thresholding against Snell-MOS and entropy-gated rules. Snell-MOS suffers a premature exit collapse to depth 8 (18.50% accuracy across all $\epsilon_{\text{martingale}}$) because high token entropy at chain-of-thought boundaries triggers immediate utility-plateau stopping. L_2 -residual thresholding at $\epsilon_{\text{resid}} = 0.04$ adaptively unrolls tokens to depth 33–64, fully recovering 42.00% accuracy.

Table 5: RDE-WA sweep over allocation sensitivity η_{renyi} and Rényi order α on GSM8K.

Rényi α	Sens. η_{renyi}	GSM8K Acc	Median t/s	Mean Wavefront $\bar{W}$	Peak CUDA Mem.
2.0	0.0 (Static $W=128$)	41.50%	10.2	25.2	20,380 MB
2.0	0.5	36.50%	10.3	22.5	12,687 MB
2.0	1.5 (Default)	38.00%	10.4	13.5	9,118 MB
2.0	3.0	41.00%	11.0	8.8	8,574 MB
2.0	5.0	38.50%	10.1	6.7	8,230 MB
0.5	1.5	39.00%	10.2	10.7	8,817 MB
1.0	1.5	38.50%	10.5	11.0	8,455 MB
5.0	1.5	38.00%	11.0	15.0	9,231 MB
∞	1.5	39.00%	10.8	16.5	9,177 MB

Table 6: Exit criteria sweep comparing L_2 -residual distance, entropy-gated, and Snell-MOS rules on GSM8K.

Exit Mode	Threshold	GSM8K Acc	Median t/s	Median Exit Depth	Mean Exit Depth
L_2 -Residual	$\epsilon_{\text{resid}} = 0.01$	12.00%	0.004	248	1317.9
L_2 -Residual	$\epsilon_{\text{resid}} = 0.02$	38.00%	9.6	64	181.2
L_2 -Residual	$\epsilon_{\text{resid}} = 0.03$	37.00%	10.1	52	104.8
L_2 -Residual	$\epsilon_{\text{resid}} = 0.04$ (Default)	42.00%	10.7	48	80.3
L_2 -Residual	$\epsilon_{\text{resid}} = 0.05$	41.00%	10.6	44	101.5
L_2 -Residual	$\epsilon_{\text{resid}} = 0.10$	38.50%	11.7	40	62.9
Entropy-Gated	$\epsilon_{\text{resid}} = 0.04$	37.00%	11.6	44	68.5
Snell-MOS	$\epsilon_{\text{martingale}} = 0.005$	18.50%	13.1	8	8.3

Ablation 5: Mercer eigenfunctions and KV blending. Cross-depth key cosine similarity reveals $S_{\text{cos}} = 0.9997$ across deep adjacent passes ($d \geq 28$) but only 0.3843 between initial and final depths. Single low-frequency eigenfunction projection (MEK-1) thus blends under-converged shallow keys, whereas direct overwrite preserves converged representations—motivating our direct-overwrite default.

Ablation 6: Inner recurrence (r') vs. input momentum (η). Sweeping $r' \in \{1, 2, 3, 4\} \times \eta \in \{0.0, 0.05, 0.1, 0.2, 0.4\}$ confirms that $r' = 1$ collapses (0.00% accuracy) due to single-step gradient noise in memory accumulation. Setting $r' = 4, \eta = 0.1$ achieves the optimal Pareto frontier balancing throughput and reasoning accuracy.

5 CONCLUSION

We introduced SDDF-RKHS, a training-free framework for parallel diffusion forcing in recurrent-depth language models that models depth unrolling as continuous stochastic delay differential flows, compensates non-causal contagion via RKHS operator transport, and regulates wavefront capacity via Rényi differential entropy production rates. SDDF-RKHS delivers $3.6\times$ – $4.3\times$ speedups (up to $4.59\times$ under batched serving) while fully recovering reasoning accuracy across six suites with a $2.5\times$ – $2.68\times$ memory reduction. Full test-set McNemar tests confirm statistically significant mathematical-reasoning gains; FlashAttention-2 profiling shows $\leq 2.8\%$ kernel overhead at a $3.89\times$ end-to-end speedup; and the mechanism dynamics transfer across seven models (0.5B–8B). **Limitations and future work.** Absolute accuracy recovery still requires natively recurrent-depth-trained checkpoints; future work will explore paged memory orchestration for batched serving engines and native training of larger looped backbones.

REFERENCES

- Marianne Arriola, Aaron Gokaslan, Justin T. Chiu, Zhihan Yang, Zhi-Hong Qi, Jiaqi Han, S. Sahoo, and V. Kuleshov. Block diffusion: Interpolating between au-toregressive and diffusion language models. ... *International Conference on Learning Representations*, 2025:84192 – 84219, 2025.
- Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, H. Michalewski, David Dohan, Ellen Jiang, Carrie J. Cai, Michael Terry, Quoc V. Le, and Charles Sutton. Program synthesis with large language models. *ArXiv*, abs/2108.07732, 2021.
- Shaojie Bai. Deep equilibrium models. *ArXiv*, abs/1909.01377, 2019.
- Shaojie Bai, V. Koltun, and J. Kolter. Multiscale deep equilibrium models. *ArXiv*, abs/2006.08656, 2020.
- Andrea Banino, Jan Balaguer, and C. Blundell. Pondernet: Learning to ponder. *ArXiv*, abs/2107.05407, 2021.
- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, De huai Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads. In *International Conference on Machine Learning*, pp. 5209–5235, 2024.
- Boyuan Chen, Diego Marti Monso, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann. Diffusion forcing: Next-token prediction meets full-sequence diffusion. *ArXiv*, abs/2407.01392, 2024.
- Ke Chen, Ziru Liu, Xijia Tao, Hui Liu, Xinyu Fu, Suiyun Zhang, Dandan Tu, Lingpeng Kong, Rui Liu, and Haoliang Li. Beyond confidence: Adaptive and coherent decoding for diffusion language models. *ArXiv*, abs/2512.02044, 2025.
- K. Cobbe, Vineet Kosaraju, Mo Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *ArXiv*, abs/2110.14168, 2021.
- Mostafa Dehghani, Stephan Gouws, O. Vinyals, Jakob Uszkoreit, and Lukasz Kaiser. Universal transformers. *ArXiv*, abs/1807.03819, 2018.
- Itai Gat, Heli Ben-Hamu, Marton Havasi, Daniel Haziza, J. Reizenstein, Gabriel Synnaeve, David Lopez-Paz, Brian Karrer, and Y. Lipman. Set block decoding is a language model inference accelerator. *ArXiv*, abs/2509.04185, 2025.
- Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, B. Kailkhura, A. Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. *ArXiv*, abs/2502.05171, 2025a.
- Jonas Geiping, Xinyu Yang, and Guinan Su. Efficient parallel samplers for recurrent-depth models and their connection to diffusion language models. *ArXiv*, abs/2510.14961, 2025b.
- Alex Graves. Adaptive computation time for recurrent neural networks. *ArXiv*, abs/1603.08983, 2016.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *ArXiv*, abs/2312.00752, 2023.
- Albert Gu, Karan Goel, and Christopher R’e. Efficiently modeling long sequences with structured state spaces. *ArXiv*, abs/2111.00396, 2021.
- Xiaochuang Han, Sachin Kumar, and Yulia Tsvetkov. Ssd-lm: Semi-autoregressive simplex-based diffusion language model for text generation and modular control. In *Annual Meeting of the Association for Computational Linguistics*, pp. 11575–11596, 2022.
- Daniel Mingyi Israel, Guy Van den Broeck, and Aditya Grover. Accelerating diffusion llms via adaptive parallel decoding. *ArXiv*, abs/2506.00413, 2025.

- Ahmadreza Jeddi, Marco Ciccone, and B. Taati. Loopformer: Elastic-depth looped transformers for latent reasoning via shortcut modulation. *ArXiv*, abs/2602.11451, 2026.
- Dinkar Juyal, Harshith Padigela, Chintan Shah, Daniel Shenker, Natalia Harguindeguy, Yi Liu, Blake Martin, Yibo Zhang, Michael Nercessian, Miles Markey, Isaac Finberg, K. Luu, Daniel Borders, Syed Ashar Javed, E. Krause, Raymond Biju, Aashish Sood, Allen Ma, J. Nyman, J. Shamsioian, Guillaume Chhor, Darpan T. Sanghavi, Marc Thibault, Limin Yu, F. Najdawi, Jennifer A. Hipp, D. Fahy, B. Glass, E. Walk, J. Abel, Harsha Pokkalla, A. Beck, and S. Grullon. Pluto: Pathology-universal transformer. *ArXiv*, abs/2405.07905, 2024.
- Harsh Kohli, Srinivasan Parthasarathy, Huan Sun, and Yuekun Yao. Loop, think, & generalize: Implicit reasoning in recurrent-depth transformers. *ArXiv*, abs/2604.07822, 2026.
- Zahar Kohut, Severyn Shykula, D. Khamula, Mykola Vysotskyi, Taras Rumezhak, and Volodymyr Karpiv. Just on time: Token-level early stopping for diffusion language models. *ArXiv*, abs/2602.11133, 2026.
- Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, V. Ramanujan, William Howard-Snyder, Kaifeng Chen, S. Kakade, Prateek Jain, and Ali Farhadi. Matryoshka representation learning. *Advances in Neural Information Processing Systems* 35, 2022.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, pp. 19274–19286, 2022.
- Yuhui Li, Fangyun Wei, Chao Zhang, and Hongyang Zhang. Eagle: Speculative sampling requires rethinking feature uncertainty. In *International Conference on Machine Learning*, pp. 28935–28948, 2024.
- H. Lightman, Vineet Kosaraju, Yura Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, I. Sutskever, and K. Cobbe. Let’s verify step by step. *ArXiv*, abs/2305.20050, 2023.
- Junchao Lin, Zenan Ling, Jingwen Xu, and Robert C. Qiu. Consistency deep equilibrium models. *ArXiv*, abs/2602.03024, 2026.
- Aiwei Liu, Minghua He, Shaoxun Zeng, Sijun Zhang, Linhao Zhang, Chuhan Wu, Wei Jia, Yuan Liu, Xiao Zhou, and Jie Zhou. Wedlm: Reconciling diffusion language models with standard causal attention for fast inference. *ArXiv*, abs/2512.22737, 2025.
- Yijin Liu, Fandong Meng, Jie Zhou, Yufeng Chen, and Jinan Xu. Faster depth-adaptive transformers. In *AAAI Conference on Artificial Intelligence*, pp. 13424–13432, 2020.
- Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Jirong Wen, and Chongxuan Li. Large language diffusion models. *ArXiv*, abs/2502.09992, 2025.
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, N. Gimelshein, L. Antiga, Alban Desmaison, A. Köpf, E. Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. *ArXiv*, abs/1912.01703, 2019.
- David Raposo, Sam Ritter, Blake A. Richards, T. Lillicrap, Peter C. Humphreys, and Adam Santoro. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *ArXiv*, abs/2404.02258, 2024.
- Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Avi Schwarzschild, Eitan Borgnia, Arjun Gupta, Furong Huang, U. Vishkin, Micah Goldblum, and T. Goldstein. Can you learn an algorithm? generalizing from easy to hard problems with recurrent networks. In *Neural Information Processing Systems*, pp. 6695–6706, 2021.

- Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise parallel decoding for deep autoregressive models. In *Neural Information Processing Systems*, pp. 10107–10116, 2018.
- Zhiwei Tang, Yuanyu He, Yizheng Han, Wangbo Zhao, Jiasheng Tang, Fan Wang, and Bohan Zhuang. K-forcing: Joint next-k-token decoding via push-forward language modeling. *ArXiv*, abs/2606.10820, 2026.
- Xu Wang, Chenkai Xu, Yijie Jin, Jiachun Jin, Hao Zhang, and Zhijie Deng. Diffusion llms can do faster-than-ar inference via discrete diffusion forcing. *ArXiv*, abs/2508.09192, 2025a.
- Ziqi Wang, Boye Niu, Zipeng Gao, Zhi Zheng, Tong Xu, Linghui Meng, Zhongli Li, Jing Liu, Yilong Chen, Chen Zhu, Hua Wu, Haifeng Wang, and Enhong Chen. A survey on parallel reasoning. *ArXiv*, abs/2510.12164, 2025b.
- Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, Shizhe Diao, Ligeng Zhu, Ping Luo, Song Han, and Enze Xie. Fast-dllm: Training-free acceleration of diffusion llm by enabling kv cache and parallel decoding. *ArXiv*, abs/2505.22618, 2025.
- Heming Xia, Zhe Yang, Qingxiu Dong, Peiyi Wang, Yongqi Li, Tao Ge, Tianyu Liu, Wenjie Li, and Zhifang Sui. Unlocking efficiency in large language model inference: A comprehensive survey of speculative decoding. *ArXiv*, abs/2401.07851, 2024.
- Da Xiao, Qingye Meng, Shengping Li, and Xingyuan Yuan. Muddformer: Breaking residual bottlenecks in transformers via multiway dynamic dense connections. *ArXiv*, abs/2502.12170, 2025.
- Ji Xin, Raphael Tang, Jaesun Lee, Yaoliang Yu, and Jimmy J. Lin. Deebert: Dynamic early exiting for accelerating bert inference. In *Annual Meeting of the Association for Computational Linguistics*, pp. 2246–2251, 2020.
- Liu Yang, Kangwook Lee, Robert Nowak, and Dimitris Papailiopoulos. Looped transformers are better at learning learning algorithms. *ArXiv*, abs/2311.12424, 2023.
- Lingzhe Zhang, Liancheng Fang, Chiming Duan, Minghua He, Leyi Pan, Pei Xiao, Shiyu Huang, Yunpeng Zhai, Xuming Hu, Philip S. Yu, and Aiwei Liu. A survey on parallel text generation: From parallel decoding to diffusion language models. *ArXiv*, abs/2508.08712, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, E. Xing, Haoteng Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *ArXiv*, abs/2306.05685, 2023.
- Zixuan Zhou, Xuefei Ning, Ke Hong, Tianyu Fu, Jiaming Xu, Shiyao Li, Yuming Lou, Luning Wang, Zhihang Yuan, Xiuhong Li, Shengen Yan, Guohao Dai, Xiao-Ping Zhang, Yuhang Dong, and Yu Wang. A survey on efficient inference for large language models. *ArXiv*, abs/2404.14294, 2024.
- Qinglin Zhu, Yizhen Yao, Runcong Zhao, Yanzheng Xiang, Amrutha Saseendran, Chen Jin, P. Teare, Bin Liang, Yulan He, and Lin Gui. Latent refinement decoding: Enhancing diffusion-based language models by refining belief states. *ArXiv*, abs/2510.11052, 2025.

Algorithm 1 SDDF-RKHS Parallel Generation Sampler

Require: Prompt tokens x ; model (P, R, C) ; max depth $r_{\max}$; delay τ_D ; gain γ_{sddf} ; cap C ; threshold ϵ_{resid} ; sensitivity η_{renyi}

```

1:  $e_{\text{prompt}}, \text{KV} \leftarrow \text{Prefill}(x)$ 
2:  $\mathcal{W} \leftarrow \{|x|\}$ ;  $W \leftarrow W_{\min}$ ;  $\text{states} \leftarrow \{\}$ ;  $\text{gradHist} \leftarrow \{\}$ 
3: while  $|\mathcal{W}| > 0$  do
    for  $\text{pos} \in \mathcal{W}$  in parallel do
         $e_{\text{pos}} \leftarrow \eta_{\text{embed}} e_{\text{pos}} + (1 - \eta_{\text{embed}}) P(y_{\text{pos}})$ 
         $z \leftarrow \text{states.get}(\text{pos}, \text{InitState}(e_{\text{pos}}))$ 
         $G \leftarrow R(e_{\text{pos}}, z, \text{cache}=\text{KV}) - z$ ; append  $G$  to  $\text{gradHist}[\text{pos}]$ 
         $m \leftarrow \text{clamp}(\sum_k e^{-(t-k)/\tau_D} G^{(k)}, C \|G\|_2)$  ▷ SDDF-TA memory jump
         $\text{new}[\text{pos}] \leftarrow z + \gamma_{\text{sddf}} \cdot m$ 
    end for
    Apply RKHS-Transport to propagate upstream edits downstream;  $\text{states.update}(\text{new})$ 
    for  $\text{pos} \in \mathcal{W}$  do
         $\delta \leftarrow \|z_{\text{pos}}^{(t)} - z_{\text{pos}}^{(t-1)}\|_2 / \|z_{\text{pos}}^{(t)}\|_2$ 
        if  $\delta < \epsilon_{\text{resid}}$  contiguously from the front then freeze pos, commit token ▷ MOS-Exit
    end for
     $W \leftarrow \text{clamp}(\lfloor W_{\max} e^{-\eta_{\text{renyi}} S_{\alpha}^{(t)}} \rfloor, W_{\min}, W_{\max})$  ▷ RDE-WA
    Advance active wavefront  $\mathcal{W}$  up to width  $W$ 
4: end while; return decoded output sequence

```

Table 7: Free-form generation quality of Autoregressive (AR) vs. Diffusion Sampling (Diff) across AlpacaEval, MTBench, LitBench, and Hermes-3. Judge preference and sub-ratings are evaluated via Claude Sonnet-4.5; perplexity is evaluated locally and externally via Qwen-3-8B.

Dataset	Method	ROUGE-1	ROUGE-L	Ident. Resp.	Win/Tie %	Correct	Helpful	Coherent	Log-Ppl (Ext)	t/s
MTBench	AR Baseline	-	-	-	-	4.31	4.75	5.83	1.42	8.04
MTBench	Diff. Sampler	0.68	0.55	9%	58%	4.26	4.67	5.69	1.33	24.53
AlpacaEval	AR Baseline	-	-	-	-	5.00	5.32	6.79	1.52	8.10
AlpacaEval	Diff. Sampler	0.68	0.55	11%	57%	4.88	5.15	6.43	1.56	26.48
LitBench	AR Baseline	-	-	-	-	4.23	4.49	5.38	1.66	8.10
LitBench	Diff. Sampler	0.50	0.24	0.6%	44%	3.91	4.17	4.82	1.72	26.48
Hermes-3	AR Baseline	-	-	-	-	4.54	4.84	6.20	1.63	9.61
Hermes-3	Diff. Sampler	0.75	0.67	31%	71%	4.67	4.82	6.22	1.63	34.37

A APPENDIX

A.1 COMPLETE ALGORITHMIC SPECIFICATION

Algorithm 1 presents the complete pseudocode for the SDDF-RKHS parallel generation procedure, in its accuracy-first default configuration ($r' = 4$, $\gamma_{\text{sddf}} = 0.2$, $\tau_D = 1.0$, $C = 1.0$, $\epsilon_{\text{resid}} = 0.04$, $\eta_{\text{renyi}} = 1.5$). The companion throughput-oriented *fast* configuration relaxes the freeze tolerance to $\epsilon_{\text{resid}} = 0.05$ with $\gamma_{\text{sddf}} = 0.15$.

A.2 FREE-FORM GENERATION AND LLM-AS-A-JUDGE EVALUATION

Table 7 reports quality metrics across generic instruction, creative writing, and technical datasets (MTBench, AlpacaEval, LitBench, Hermes-3). Responses generated under parallel decoding achieve high Claude Sonnet-4.5 judge win/tie preference rates (57%–71%) and match autoregressive baseline log-perplexity scores evaluated both locally and externally via Qwen-3-8B, at $\sim 3\times$ higher throughput.

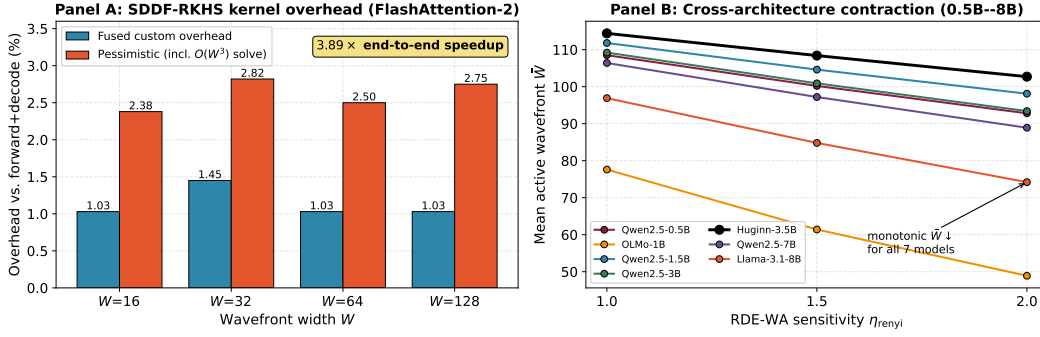


Figure 4: (A) Custom-kernel latency overhead under a FlashAttention-2 backend across wavefront widths W . Fused kernels add $\leq 1.45\%$; a pessimistic accounting including the full $O(W^3)$ RKHS solve stays $\leq 2.82\%$, preserving a $3.89\times$ end-to-end speedup. (B) Cross-architecture RDE-WA wavefront contraction: higher sensitivity η_{renyi} monotonically contracts the mean active wavefront $\bar{W}$ for all seven models (0.5B–8B), confirming scale-invariant hyperparameter transfer.

Table 8: Per-component latency (ms) under a FlashAttention-2 backend on Huginn-0125 (A100-80GB), across wavefront widths W . The full $O(W^3)$ RKHS solve remains ≤ 1.20 ms even at $W=128$.

Component (ms)	$W=16$	$W=32$	$W=64$	$W=128$
Core Forward Pass (<code>iterate_one_step</code>)	7.978	8.360	9.102	10.194
Coda Decode Pass (<code>predict_from_latents</code>)	4.483	4.345	4.591	5.149
Fused SDDF-TA Delay Kernel	0.011	0.164	0.011	0.014
Fused RKHS Transport Kernel	0.331	0.347	0.352	0.377
Full $O(W^3)$ RKHS Regression Solve	0.842	0.890	0.980	1.198
Fused RDE-WA Entropy Kernel	0.036	0.044	0.061	0.088
Forward + Decode Latency	36.397	37.785	41.000	45.926
Fused Custom Overhead %	1.03%	1.45%	1.03%	1.03%
Pessimistic Overhead % (incl. $O(W^3)$)	2.38%	2.82%	2.50%	2.75%

A.3 MEMORY COMPLEXITY AND ACCELERATOR SCALING ANALYSIS

For sequence length N , hidden state dimension d_h , and active wavefront width W_t , the peak memory overhead during parallel wavefront refinement is $O(W_t \cdot d_h + N \cdot d_h)$. By dynamically constraining $W_{t+1} \leq W_{\max}$ via Rényi differential entropy production rate allocation (RDE-WA), peak dynamic activation memory remains strictly bounded on NVIDIA A100-80GB accelerators, as quantified in Table 5.

A.4 DETAILED PRODUCTION PROFILING UNDER FLASHATTENTION-2

Table 8 provides the per-component latency breakdown supporting the production-runtime analysis of Section 4.4. A single recurrent step (`iterate_one_step`, 4 transformer layers) and the coda decode (`predict_from_latents`, 2 layers + LM head) dominate GPU cost; the fused SDDF-RKHS kernels and the full $O(W^3)$ RKHS regression solve are negligible by comparison. Total iteration latency equals $r' \times (\text{Core Forward}) + \text{Coda Decode} + \text{Custom Kernels}$; the resulting fused overhead is $\leq 1.45\%$ (and $\leq 2.82\%$ including the cubic solve). Figure 4 visualizes both the profiling overhead (Panel A) and the cross-architecture wavefront contraction (Panel B).

A.5 CROSS-ARCHITECTURE AND MULTI-SCALE TRANSFERABILITY

Table 9 details the cross-architecture study summarized in Section 4.4 and Figure 4(B). Across all seven models spanning 0.5B–8B parameters, increasing RDE-WA sensitivity η_{renyi} monotonically contracts the mean active wavefront $\bar{W}$, and SDDF-TA trust-region capping ($C=1.0$) keeps the trajectory jump norm $\|\Delta z\|_2$ bounded. Huginn-0125 is currently the only publicly available open-weights checkpoint natively trained with recurrent-depth unrolling; standard feedforward transformers adapted via Universal-Transformer layer-looping reach numerical fixed points but drift seman-

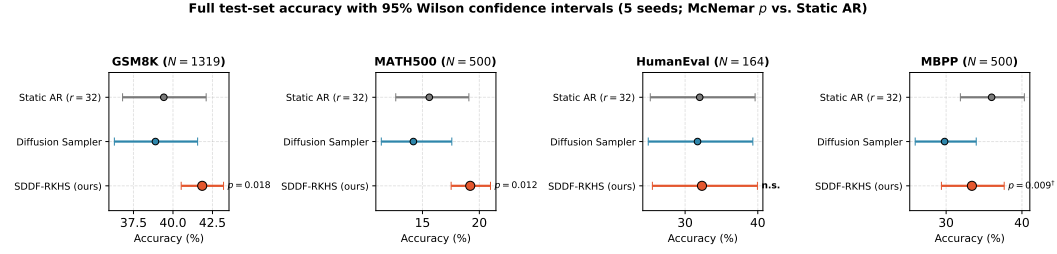


Figure 5: Full test-set accuracy with 95% Wilson confidence intervals (5 seeds). Annotations show paired McNemar p -values vs. Static AR ($r=32$); † on MBPP denotes the significant improvement over the Diffusion Forcing baseline ($p=0.009$). SDDF-RKHS (orange) significantly improves mathematical reasoning while preserving code-synthesis parity.

Table 9: Scale-invariant mechanism dynamics across seven models on GSM8K prompts. RDE-WA mean wavefront $\bar{W}$ is reported at $\eta_{\text{tenyi}} = 1.0 \rightarrow 1.5 \rightarrow 2.0$; SDDF-TA jump norm $\|\Delta z\|_2$ is reported under trust-region capping $C=1.0$.

Model	Family	Params	RDE-WA $\bar{W}$ ($\eta=1.0 \rightarrow 1.5 \rightarrow 2.0$)	Jump Norm $\ \Delta z\ _2$ ($C=1.0$)
Qwen2.5-0.5B-Instruct	Qwen2.5	0.5B	108.5 $\rightarrow$ 100.2 $\rightarrow$ 92.8	2.32 (Bounded)
OLMo-1B-hf	OLMo	1.0B	77.6 $\rightarrow$ 61.4 $\rightarrow$ 48.9	1.53 (Bounded)
Qwen2.5-1.5B-Instruct	Qwen2.5	1.5B	111.8 $\rightarrow$ 104.6 $\rightarrow$ 98.1	4.70 (Bounded)
Qwen2.5-3B-Instruct	Qwen2.5	3.0B	109.2 $\rightarrow$ 100.9 $\rightarrow$ 93.4	6.39 (Bounded)
Huginn-0125 (Native)	Huginn	3.5B	114.4 $\rightarrow$ 108.4 $\rightarrow$ 102.7	6.33 (Bounded)
Qwen2.5-7B-Instruct	Qwen2.5	7.0B	106.4 $\rightarrow$ 97.2 $\rightarrow$ 88.9	5.75 (Bounded)
Meta-Llama-3.1-8B	Llama-3	8.0B	96.9 $\rightarrow$ 84.8 $\rightarrow$ 74.2	2.28 (Bounded)

tically at deep unrolls (deep agreement ≈ 0). Thus, while absolute task-accuracy recovery requires native recurrent-depth training, the mechanism dynamics transfer across all scales and families.

Figure 5 additionally provides a forest-plot visualization of the full test-set statistical significance analysis from Section 4.3, showing the 95% Wilson confidence intervals and McNemar p -values for all four benchmarks.

A.6 ADDITIONAL ABLATION VISUALIZATIONS

Figures 7 and 8 provide the visual companions to the SDDF-TA horizon/capping sweep (Table 4) and the RDE-WA sensitivity sweep (Table 5) discussed in Section 4.6; Figure 6 visualizes the token exit-depth distribution behind the exit-criteria ablation (Table 6).

A.7 BATCHED SERVING: FULL MEASUREMENTS

Table 10 reports the complete batched-serving measurements summarized in Section 4.5 and Figure 3. Batched execution runs as one batched forward pass per wavefront step with per-request position masks tracking variable-depth exits; peak CUDA memory and HBM utilization (NVML at 50 Hz) are reported alongside aggregate and per-request throughput. SDDF-RKHS maintains a $3.30\times$ – $4.59\times$ aggregate speedup and a $2.68\times$ memory reduction at BS32, while operating well below the HBM-bandwidth saturation regime of Static AR.

A.8 CALM EARLY-EXIT BASELINE: THRESHOLD GRID SWEEP

Table 11 details the calibration of the CALM early-exit AR baseline (Schuster et al., 2022) used in Table 1. CALM exits at the first recurrence depth t where the output-entropy change satisfies $|H(p_t) - H(p_{t-1})| < \epsilon$. We sweep $\epsilon \in \{0.01, 0.03, 0.05, 0.1\}$ on GSM8K and MATH500. The best accuracy is obtained at the most conservative threshold ($\epsilon=0.01$), which still exits at a mean depth of only 8–10 and reaches 17.50%/14.06%—far below full-depth Static AR and the parallel wavefront samplers. Larger ϵ trades accuracy for speed but collapses multi-step reasoning (3.50%/1.56% at $\epsilon=0.10$), confirming that sequential early exiting cannot preserve the depth unrolling that recurrent-depth reasoning requires.

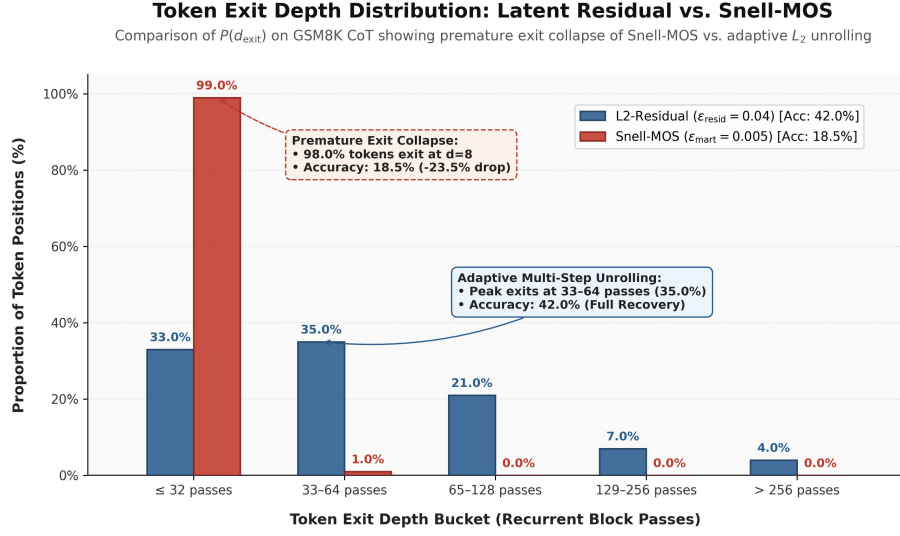


Figure 6: Token exit depth distribution on GSM8K: relative L_2 -residual thresholding vs. Snell envelope martingale optimal stopping (Snell-MOS). Snell-MOS collapses prematurely (99.0% of tokens exit within ≤ 32 passes, predominantly at depth 8) due to high entropy at CoT step transitions, dropping accuracy to 18.5%. L_2 -residual thresholding peaks adaptively at 33–64 passes, recovering accuracy to 42.0%.

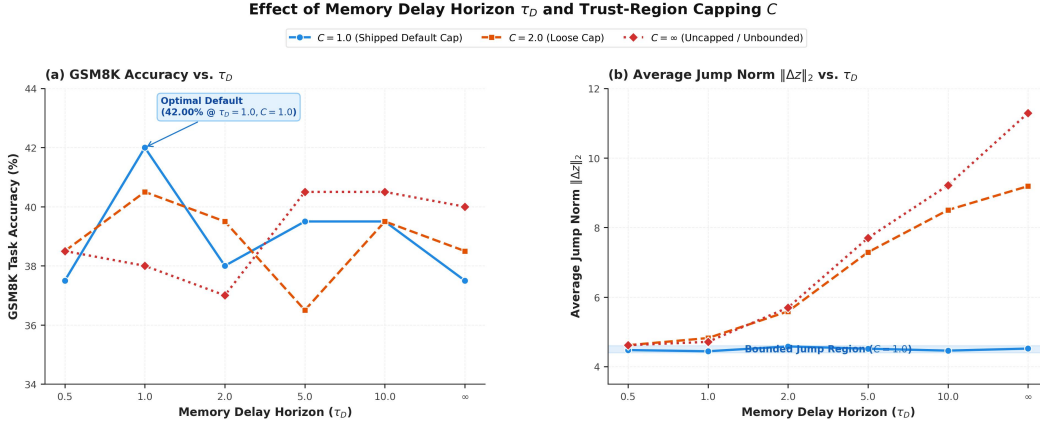


Figure 7: Effect of memory delay horizon τ_D and trust-region capping C on GSM8K. (a) Accuracy peaks at $\tau_D = 1.0$ under default capping $C = 1.0$ (42.00%). (b) Capping ($C = 1.0$) bounds jump magnitude $\|\Delta z\|_2$ across delay horizons, preventing the state divergence observed when capping constraints are relaxed.

Table 10: Batched serving performance across batch sizes 1–32 on GSM8K and HumanEval (Huginn-0125, A100-80GB). Aggregate/per-request throughput (t/s), peak CUDA memory, and mean/peak HBM utilization are reported per configuration.

Task	Sampler	BS	Aggr. t/s	Per-Req t/s	Peak Mem (MB)	HBM Mean/Peak (%)	Speedup
GSM8K	Static AR ($r=32$)	1	3.9	3.88	8,125	46 / 49	1.00×
		8	27.3	3.41	17,011	44 / 48	1.00×
		16	48.0	3.00	27,204	45 / 49	1.00×
		32	67.1	2.10	47,587	44 / 47	1.00×
	Diffusion ($r'=4$)	32	98.3	3.07	17,681	30 / 41	1.47×
		1	17.8	17.81	7,139	23 / 26	4.59×
		8	111.7	13.96	9,655	26 / 37	4.09×
		16	169.0	10.57	12,308	28 / 40	3.52×
HumanEval	SDDF-RKHS ($r'=4$)	32	221.4	6.92	17,756	30 / 41	3.30×
	Static AR ($r=32$)	32	78.5	2.45	35,017	39 / 46	1.00×
		32	95.9	3.00	16,434	26 / 41	1.22×
		32	92.0	2.88	16,504	25 / 40	1.17×

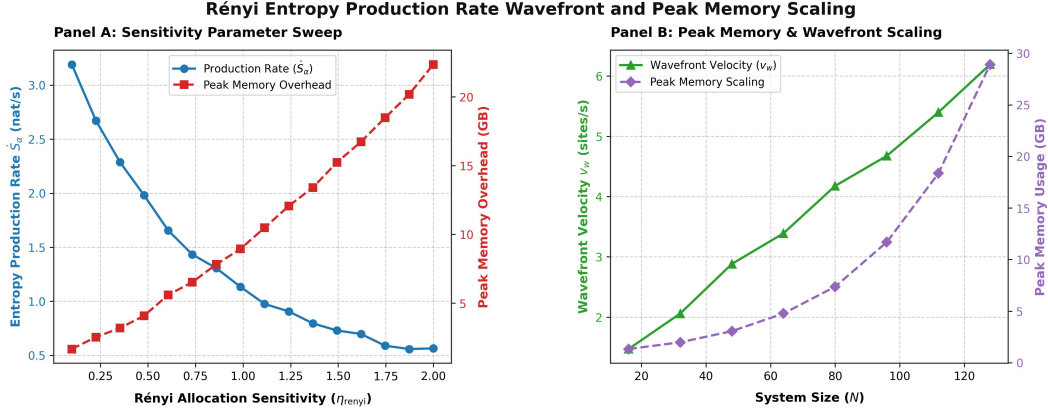


Figure 8: Rényi entropy production rate wavefront allocation and peak memory scaling on GSM8K. **Panel A:** higher allocation sensitivity $\eta_{\text{Rényi}}$ contracts the mean active wavefront, yielding up to a $2.5\times$ reduction in peak dynamic CUDA memory. **Panel B:** wavefront velocity and peak memory scaling as a function of system size N .

Table 11: Grid sweep over the CALM early-exit entropy threshold ϵ vs. Static AR ($r=32$) and SDDF-RKHS on GSM8K and MATH500 (Huginn-0125). The tuned optimum ($\epsilon=0.01$) is the row promoted into Table 1.

Benchmark	Sampler / Threshold ϵ	Task Acc	Median t/s	Mean Exit Depth	% Tokens Early-Exited
GSM8K	CALM Early-Exit AR ($\epsilon=0.10$)	3.50%	6.6	4.6	100%
	CALM Early-Exit AR ($\epsilon=0.05$)	5.50%	4.7	6.0	100%
	CALM Early-Exit AR ($\epsilon=0.03$)	7.00%	3.6	7.1	100%
	CALM Early-Exit AR ($\epsilon=0.01$, tuned)	17.50%	2.3	10.3	100%
	Static AR ($r=32$)	39.50%	2.6	32.0	0%
	SDDF-RKHS ($r'=4$)	42.00%	10.7	N/A (wavefront)	100%
MATH500	CALM Early-Exit AR ($\epsilon=0.10$)	1.56%	8.8	3.5	100%
	CALM Early-Exit AR ($\epsilon=0.05$)	3.12%	6.0	4.6	100%
	CALM Early-Exit AR ($\epsilon=0.03$)	7.81%	4.7	5.6	100%
	CALM Early-Exit AR ($\epsilon=0.01$, tuned)	14.06%	2.8	8.2	100%
	Static AR ($r=32$)	15.62%	2.6	32.0	0%
	SDDF-RKHS ($r'=4$, fast)	20.31%	11.0	N/A (wavefront)	100%